



Infineon BCR 450 LED driver IC

circuit operating in a hot environment. These solutions also improve the performance with electronic transformers by reducing flickering.

Piero Bianco, business manager-Battery, Display and Industrial Power business unit, Maxim says, "Street-lighting and outdoor lighting is another strongly growing market. In this field, the dominant driver solution is an off-the-shelf AC-DC converter that generates a DC intermediate voltage and then have LED drivers running off this voltage. The main requirement for these drivers is probably to have a low external component count and low solution cost."

In automotive applications, LEDs are already being used for rear lamps and now moving into daytime-running lights and position and high-/low-beam lights. LEDs for indicator and illumination lighting, turn signals, brake lights, map reading lights and headlights too are expected to see rapid growth.

One of the major applications particularly relevant for India is solar-powered LED lamps and lanterns. Bianco explains, "This market requires DC-input drivers that must be robust and easy to design with low external component count and solution cost."

Why do you need an LED driver?

An LED driver provides constant-current supply to the LED when there are variations in the supply voltage. Keong shares, "The purpose of the LED driver IC is to improve the thermal efficiency and reliability of LED lighting products." It also provides smart functions such as PWM-based dimming control and protection; e.g., electronic circuitry protection in the event of LED short-circuit or sudden surge in input or output voltage. Implementing these circuits on silicon ultimately contributes to reducing the environmental impacts.

LED drivers are needed to regulate and step-down the supply voltage depending on the application. Bianco explains, "For 120V or 230V input application, a lamp without drivers must have many LEDs in series (each LED has a voltage drop of the order of 3V). Having many LEDs can be okay for certain kind of lamps, but, in general, for optical reasons a directional lamp with good light quality must have fewer high-power LEDs." This requires special drivers to supply the necessary voltage and current.

Designing a dimming LED using triac dimmers too requires an LED driver. Bianco says, "The correct driv-

ers can make an LED lamp dimmable, while LED lamps without drivers are not dimmable with triac dimmers."

Challenges in LED application design

LEDs are current-sensitive components and therefore the target applications need to be carefully designed. Keong shares, "Maintaining uniform brightness (current accuracy), high thermal efficiency, luminary diffuser design to allow uniform light emission by LEDs, heat dissipation management, and voltage and current protection functions are some of the design challenges faced by LED application designers."

He adds, "In the last few years, there has been a significant progress in LED drivers' ability to provide higher current accuracy by implementing both current- and voltage-feedback mechanisms. There has been improvement in the optical design of diffusers and design of heat-sinks."

Since LEDs basically need a DC current, the LED driver can be a DC-DC converter or AC-DC converter. According to Bianco, "For DC input drivers, the challenges are similar to the design of DC-DC converters, like compensating the control loop and solving noise problems, of course at the lowest cost. The additional requirement can be thermal management of LEDs; LEDs dissipate much energy, and smart solutions may be needed to keep them cool enough."

The design of AC input drivers is more complex. Bianco explains, "AC input power supplies are more complex by themselves because of the higher voltages that are present, higher noise of fly-back converters, etc. In addition, a lot of work is needed to fit in the requirements for electromagnetic interference (EMI), total harmonic distortion (THD), power factor correction (PFC) and, in particular, dimmability and compatibility with electronic transformers."

EMI problem is often solved with the correct board-level design. In order to solve issues of dimmability and compatibility with transformers,